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These solutions are not the only “perfect” or “10/10” solutions to be followed as a model. They are examples of ways to solve these problems. There are many many alternative ways to solve these problems. We hope that these example solutions help you check your work.

Problems

Q1. (a) Double addition of real numbers is given by the formula:

$$a \boxplus b = 2a + 2b$$

where $+$ is the usual addition of real numbers.

Is $\boxplus$ associative?

Solution: No! Note that

$$(a \boxplus b) \boxplus c = (2a + 2b) \boxplus c = 2(2a + 2b) + 2c = 4a + 4b + 2c.$$

Nevertheless,

$$a \boxplus (b \boxplus c) = a \boxplus (2b + 2c) = 2a + 2(2b + 2c) = 2a + 4b + 4c.$$

These quantities are, in general, different. Consider, for example, the case in which $a = b = 1$ and $c = 0$.

(b) Halfsy addition of real numbers is given by the formula:

$$a \boxplus b = 2a + b$$

where $+$ is the usual addition of real numbers.

Is $\boxplus$ associative?

Solution: No! Note that

$$(a \boxplus b) \boxplus c = (2a + b) \boxplus c = 2(2a + b) + c = 4a + 2b + c.$$

Nevertheless,

$$a \boxplus (b \boxplus c) = a \boxplus (2b + c) = 2a + 2b + c.$$

These quantities are, in general, different. Consider, for example, the case in which $a = b = 1$ and $c = 0$.

Q2. Complicated multiplication of pairs of real numbers is given by the formula:

$$(a, b) \boxdot (c, d) = (ac - bd, ad + bc)$$

where addition, subtraction, and multiplication are the usual operations on real numbers.

(a) Determine (with proof) whether $\boxdot$ is associative.

Solution: It is associative. Note that

$$\begin{aligned} ((a, b) \boxdot (c, d)) \boxdot (e, f) &= (ac - bd, ad + bc) \boxdot (e, f) \\ &= ((ac - bd)e - (ad + bc)f, (ac - bd)f + (ad + bc)e) \\ &= (a(ce - df) - b(de + cf), a(de + cf) + b(ce - df)) \\ &= (a, b) \boxdot (ce - df, de + cf) \\ &= (a, b) \boxdot ((c, d) \boxdot (e, f)). \end{aligned}$$

(b) Determine (with proof) whether $\boxdot$ is commutative.

Solution: It is commutative. Note that

$$\begin{aligned} (a, b) \boxdot (c, d) &= (ac - bd, ad + bc) \\ &= (ca - db, da + cb) = (c, d) \boxdot (a, b). \end{aligned}$$

Q3. The axioms of a vector space require the existence of an additive inverse for each element.

For all $\mathbf{v} \in V$ there is $\mathbf{w} \in V$ such that $\mathbf{v} + \mathbf{w} = \mathbf{0}$.
We say: $\mathbf{w}$ is the additive inverse of $\mathbf{v}$.

Prove the following:

(a) Additive inverses are unique.

Solution: Let's assume that $\mathbf{w}_1$ and $\mathbf{w}_2$ are additive inverses for the vector $\mathbf{v} \in V$, this is, that $\mathbf{v} + \mathbf{w}_1 = \mathbf{v} + \mathbf{w}_2 = \mathbf{0}$. We want to conclude that $\mathbf{w}_1 = \mathbf{w}_2$, implying the uniqueness of the additive inverse of $\mathbf{v}$. To do this, note that

$$\mathbf{w}_1 = \mathbf{w}_1 + \mathbf{0} = \mathbf{w}_1 + (\mathbf{v} + \mathbf{w}_2) = (\mathbf{w}_1 + \mathbf{v}) + \mathbf{w}_2 = \mathbf{0} + \mathbf{w}_2 = \mathbf{w}_2.$$

(b) The additive inverse of the additive inverse of $\mathbf{v}$ is $\mathbf{v}$.

It might help to read the question as: (The additive inverse of (the additive inverse of $\mathbf{v}$)) is $\mathbf{v}$.

Solution: Just note that if $\mathbf{w}$ is an additive inverse for $\mathbf{v} \in V$, then $\mathbf{w} + \mathbf{v} = \mathbf{0}$, which can be rephrased as saying that $\mathbf{v}$ is an additive inverse for $\mathbf{w}$. Since additive inverses are unique, by the first part of this problem, then $\mathbf{v}$ is the additive inverse of its additive inverse.

Q4. Consider $\mathbb{R}^+ = \{x : x > 0\}$. We define the following operations:

$$x \boxplus y = xy \quad c \boxdot x = x^c \text{ for } c \in \mathbb{R}$$

Show that $V = (\mathbb{R}^+, \boxplus, \boxdot)$ is an $\mathbb{R}$ -vector space.

Solution:

1. Since the multiplication of real numbers is associative, we have that

$$(x \boxplus y) \boxplus z = xy \boxplus z = xyz = x \boxplus yz = x \boxplus (y \boxplus z).$$

2. Since the multiplication of real numbers is commutative, we have that

$$x \boxplus y = xy = yx = y \boxplus x.$$

3. Note that the real number 1 satisfies that for every real number x ,

$$x \boxplus 1 = x \cdot 1 = x,$$

so 1 is the additive identity in this context.

4. If x is a positive real, then also x^{-1} is positive, and also

$$x \boxplus x^{-1} = xx^{-1} = 1,$$

and therefore x^{-1} is the additive inverse of x .

5. To check the first distributivity axiom, let $c, x, y \in \mathbb{R}$, and note that

$$c \boxdot (x \boxplus y) = c \boxdot xy = (xy)^c = x^c y^c = (c \boxdot x) \boxplus (c \boxdot y).$$

6. For the second distributivity axiom, let $c, d, x \in \mathbb{R}$, and note that

$$(c + d) \boxdot x = x^{c+d} = x^c \cdot x^d = (c \boxdot x) \boxplus (d \boxdot x).$$

7. Now, if $c, d, x \in \mathbb{R}$, we have that

$$c \boxdot (d \boxdot x) = c \boxdot (x^d) = (x^d)^c = x^{cd} = (cd) \boxdot x.$$

8. Finally, for every positive real number x , we have that

$$1 \boxdot x = x^1 = x.$$

Thus, we can conclude that the structure defined above is indeed a vector space.

Q5. Consider any vector space V .

Let $\mathbf{x}, \mathbf{y}, \mathbf{z}$ be elements of V and a, b be scalars.

(a) Prove: If $\mathbf{x} + \mathbf{y} = \mathbf{x} + \mathbf{z}$ then $\mathbf{y} = \mathbf{z}$.

Solution: Let $-\mathbf{x}$ be the additive inverse of $\mathbf{x}$, and note that

$$\mathbf{y} = \mathbf{y} + (\mathbf{x} - \mathbf{x}) = (\mathbf{x} + \mathbf{y}) - \mathbf{x} = (\mathbf{x} + \mathbf{z}) - \mathbf{x} = \mathbf{z} + (\mathbf{x} - \mathbf{x}) = \mathbf{z}.$$

(b) Prove: $(a + b)(\mathbf{x} + \mathbf{y}) = a\mathbf{x} + b\mathbf{x} + a\mathbf{y} + b\mathbf{y}$.

Solution: For this problem, we need to use the two distributivity axioms:

First, by distributing the scalar $a + b$ over the vector addition $\mathbf{x} + \mathbf{y}$, we have that

$$(a + b)(\mathbf{x} + \mathbf{y}) = (a + b)\mathbf{x} + (a + b)\mathbf{y}.$$

Now, distributing the vectors $\mathbf{x}$ and $\mathbf{y}$ over the scalar addition, we obtain that

$$(a + b)\mathbf{x} + (a + b)\mathbf{y} = a\mathbf{x} + b\mathbf{x} + a\mathbf{y} + b\mathbf{y},$$

as desired.

Q6. Consider polynomials with real coefficients and degree at most n .

$$\mathbf{P}_n(\mathbb{R}) = \{a_0 + a_1x + a_2x^2 + \cdots + a_nx^n : a_i \in \mathbb{R}\}$$

Prove that $\mathbf{P}_n(\mathbb{R})$ is a vector space.

Solution: Let us fix three arbitrary polynomials in $\mathbf{P}_n(\mathbb{R})$,

- $p(x) = a_0 + a_1x + \cdots + a_nx^n$
- $q(x) = b_0 + b_1x + \cdots + b_nx^n$
- $r(x) = c_0 + c_1x + \cdots + c_nx^n$

1. Note that

$$\begin{aligned} (p(x) + q(x)) + r(x) &= (a_0 + a_1x + \cdots + a_nx^n + b_0 + b_1x + \cdots + b_nx^n) + c_0 + c_1x + \cdots + c_nx^n \\ &= (a_0 + b_0 + c_0) + (a_1 + b_1 + c_1)x + \cdots + (a_n + b_n + c_n)x^n \\ &= a_0 + a_1x + \cdots + a_nx^n + (b_0 + b_1x + \cdots + b_nx^n + c_0 + c_1x + \cdots + c_nx^n) \\ &= p(x) + (q(x) + r(x)). \end{aligned}$$

2. Also,

$$\begin{aligned} p(x) + q(x) &= a_0 + a_1x + \cdots + a_nx^n + b_0 + b_1x + \cdots + b_nx^n \\ &= (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n \\ &= (b_0 + a_0) + (b_1 + a_1)x + \cdots + (b_n + a_n)x^n \\ &= b_0 + b_1x + \cdots + b_nx^n + a_0 + a_1x + \cdots + a_nx^n \\ &= q(x) + p(x). \end{aligned}$$

3. Note that the zero polynomial, that is, the constantly zero function, has degree at most n and satisfies that $p(x) + \mathbf{0} = p(x)$ for every polynomial $p(x)$. Therefore, the constantly zero polynomial is the additive identity in $\mathbf{P}_n(\mathbb{R})$.

4. Given a polynomial

$$p(x) = a_0 + a_1x + \cdots + a_nx^n,$$

consider the polynomial $s(x) = -a_0 - a_1x - \cdots - a_nx^n$. Then

$$\begin{aligned} p(x) + s(x) &= a_0 + a_1x + \cdots + a_nx^n + (-a_0 - a_1x - \cdots - a_nx^n) \\ &= (a_0 - a_0) + (a_1 - a_1)x + \cdots + (a_n - a_n)x^n \\ &= \mathbf{0}. \end{aligned}$$

5. Given a real number d , note that

$$\begin{aligned} d(p(x) + q(x)) &= d(a_0 + a_1x + \cdots + a_nx^n + b_0 + b_1x + \cdots + b_nx^n) \\ &= d[(a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n] \\ &= (da_0 + db_0) + (da_1 + db_1)x + \cdots + (da_n + db_n)x^n \\ &= da_0 + da_1x + \cdots + da_nx^n + db_0 + db_1x + \cdots + db_nx^n \\ &= dp(x) + dq(x). \end{aligned}$$

6. For the second distributivity axiom, let $d, e \in \mathbb{R}$, and note that

$$\begin{aligned} (d + e)p(x) &= (d + e)(a_0 + a_1x + \cdots + a_nx^n) \\ &= (d + e)a_0 + (d + e)a_1x + \cdots + (d + e)a_nx^n \\ &= da_0 + da_1x + \cdots + da_nx^n + ea_0 + ea_1x + \cdots + ea_nx^n \\ &= dp(x) + ep(x). \end{aligned}$$

7. Given again two real numbers d and e , we have that

$$\begin{aligned} d(ep(x)) &= d(ea_0 + ea_1x + \cdots + ea_nx^n) \\ &= dea_0 + dea_1x + \cdots + dea_nx^n \\ &= (de)(a_0 + a_1x + \cdots + a_nx^n) \\ &= (de)p(x). \end{aligned}$$

8. Finally, note that

$$1 \cdot p(x) = 1 \cdot (a_0 + a_1x + \cdots + a_nx^n) = a_0 + a_1x + \cdots + a_nx^n = p(x).$$

Thus, we can conclude that $\mathbf{P}_n(\mathbb{R})$ endowed with the usual operations is a vector space.